

1. Reverse a String

Buggy Code:

```
string s="hello", rev="";
for(int i=0;i<=s.length();i++)
    rev=s[i]+rev;
cout<<rev;
```

Correct Code:

```
string s="hello", rev="";
for(int i=s.length()-1;i>=0;i--)
    rev+=s[i];
cout<<rev;
```

2. String Palindrome

Buggy Code:

```
string s="madam", r=s;
reverse(s.begin(),s.end());
if(s==r) cout<<"Palindrome";
```

Correct Code:

```
string s="madam";
int i=0,j=s.length()-1; bool pal=true;
while(i<j){
    if(s[i++]!=s[j--]){ pal=false; break;}
}
cout<<(pal?"Palindrome":"Not Palindrome");
```

3. Count Vowels

Buggy Code:

```
string s="hello"; int count=0;
for(int i=0;i<s.length();i++)
    if(s[i]=='a' || 'e' || 'i' || 'o' || 'u') count++;
cout<<count;
```

Correct Code:

```
string s="hello"; int count=0;
for(char c:s)
    if(c=='a' || c=='e' || c=='i' || c=='o' || c=='u') count++;
cout<<count;
```

4. Largest Element in Array

Buggy Code:

```
int a[]={3,7,2,9};
int max=0;
for(int i=0;i<4;i++)
    if(a[i]>max) max=a[i];
cout<<max;
```

Correct Code:

```
int a[]={3,7,2,9};
int max=a[0];
```

```
for(int i=1;i<4;i++)
    if(a[i]>max) max=a[i];
cout<<max;
```

5. Second Largest Element

Buggy Code:

```
int a[]={4,1,7,3};
int first=0,second=0;
for(int i=0;i<4;i++)
    if(a[i]>first) second=first,first=a[i];
cout<<second;
```

Correct Code:

```
int a[]={4,1,7,3};
int first=INT_MIN,second=INT_MIN;
for(int i=0;i<4;i++){
    if(a[i]>first){ second=first; first=a[i];}
    else if(a[i]>second && a[i]!=first) second=a[i];
}
cout<<second;
```

6. Sort Array Ascending

Buggy Code:

```
int a[]={3,1,2};
for(int i=0;i<3;i++)
    for(int j=i+1;j<3;j++)
        if(a[i]<a[j]) swap(a[i],a[j]);
```

Correct Code:

```
int a[]={3,1,2};
for(int i=0;i<3;i++)
    for(int j=i+1;j<3;j++)
        if(a[i]>a[j]) swap(a[i],a[j]);
```

7. Remove Duplicates

Buggy Code:

```
int a[]={1,2,2,3};
for(int i=0;i<4;i++)
    cout<<a[i]<<" ";
```

Correct Code:

```
int a[]={1,2,2,3};
set<int> s(a,a+4);
for(int x:s) cout<<x<<" ";
```

8. Character Frequency

Buggy Code:

```
string s="aab";
map<char,int> m;
```

```
for(int i=0;i<=s.length();i++)
    m[s[i]]++;
```

Correct Code:

```
string s="aab";
map<char,int> m;
for(char c:s) m[c]++;
for(auto p:m) cout<<p.first<<": "<<p.second<<" ";
```

9. Sum of Array Elements

Buggy Code:

```
int a[]={1,2,3},sum;
for(int i=0;i<3;i++)
    sum+=a[i];
cout<<sum;
```

Correct Code:

```
int a[]={1,2,3},sum=0;
for(int i=0;i<3;i++)
    sum+=a[i];
cout<<sum;
```

10. Merge Two Arrays

Buggy Code:

```
int a[]={1,3},b[]={2,4};
int c[4];
for(int i=0;i<2;i++) c[i]=a[i];
for(int i=0;i<2;i++) c[i]=b[i];
```

Correct Code:

```
int a[]={1,3},b[]={2,4};
int c[4],k=0;
for(int i=0;i<2;i++) c[k++]=a[i];
for(int i=0;i<2;i++) c[k++]=b[i];
for(int i=0;i<4;i++) cout<<c[i]<<" ";
```